Yaobin Yang, Ph.D.

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Education

09/2020 – 12/2024 **Ph.D., Civil Engineering, University of California, Berkeley** in USA.
Thesis title: *Discontinuous Galerkin Finite Element Analysis and Experimental Sensors Innovations for Unsaturated Soil.*
Minors: Structural Engineering, Applied Mathematics
Advisor: Prof. Kenichi Soga

09/2019 – 09/2020 **M.Sc., Civil Engineering, University of California, Berkeley** in USA.

07/2015 – 09/2019 **B.E. Civil Engineering, Tongji University** in China.
Minor: Applied Mathematics

Working and Visiting Experience

01/2025 – present **Postdoctoral Researcher.** University of California, Berkeley.

09/2020 – 09/2024 **Graduate Student Researcher,** University of California, Berkeley.

01/2019 – 05/2019 **Graduate Student Instructor,** University of California, Berkeley.

05/2019 – 09/2019 **Intern.** Shanghai Tunnel Eng Co.

05/2018 – 09/2018 **Intern.** China Railway Design Corporation.

10/2023 – 01/2024 **Visiting Scientist.** Rothamsted Research.

09/2022 – 12/2025 **Research Affiliate.** Lawrence Berkeley National Laboratory.

09/2018 – 05/2019 **Visiting Student.** University of California, Berkeley.

Research Interests

Infrastructure	Structural health monitoring, Pipe leak detection, Geotechnical structures.
Sensing Technology	Fiber-optics Sensing, Shape & Tactile Sensing.
Numerical Computation	FEM, Multi-physics, Discontinuous Galerkin, Unsaturated soil.
Geophysics	Inversion, Distributed acoustic sensing.
Optimization	PDECO, Topology optimization, Optimal control.
Robotics	Soft robotics, Construction robotics, 3D printing.

Research Publications

Journal Articles

1. G.-B. Shin, **Y. Yang***, L. Pelecanos, F. A. Mevoli, A. Yeskoo, T. Shantz, and K. Soga, "Distributed fiber optic sensing for monitoring piles under eccentric load," *Journal of Geotechnical and Geoenvironmental Engineering*, 2026, doi:10.1061/JGGEFK/GTENG-14998.

2. **Y. Yang**, G.-B. Shin*, L. Pelecanos, C.-C. Wang, L. Luo, and K. Soga, "Inference of pile capacity from distributed strain sensing via pde-constrained optimization," *Computers and Geotechnics*, vol. 196, p. 108 131, 2026, ISSN: 0266-352X.

3. **Y. Yang***, L. Luo, M. Jasiak, C.-C. Wang, S.-H. Chiu, and K. Soga, "Cosserat rod-based modeling and stochastic analysis for distributed fiber optics shape-sensing," *Mechanical Systems and Signal Processing*, vol. 241, p. 113 457, 2025.

4. F. Cheng, X. Sun*, Y. Li*, X. Ju, **Y. Yang**, X. Liu, W. Liu, M. Yang, and Y. Song, "Numerical analysis of coupled thermal-hydro-chemo-mechanical (thcm) behavior to joint production of marine gas hydrate and shallow gas," *Energy*, vol. 281, p. 128 224, 2023.

Thesis

1. **Y. Yang**, *Discontinuous galerkin finite element analysis and experimental sensors innovations for unsaturated soil*, 2024.

Conference Proceedings

1. S. K. Sinha*, J. Yang, **Y. Yang**, A. Nigh, and K. Soga, "Thermal response test of geothermal boreholes using distributed fiber optic sensing," in *Geo-Congress 2024*, pp. 493–503.
2. **Y. Yang***, L. Luo, K. Soga, and X. Sun*, "Fiber-optic miniature pore pressure probe – design, fabrication and application," in *ASCE Conference 2027*, Accepted Abstract, Paper Forthcoming. Abstract ID: ASCE2027-288, 2027.
3. T. H. Illangasekare*, A. M. Ilie, **Y. Yang**, K. Soga, W. R. Whalley, and A. Trautz, "Intermediate-scale testing of a spatially distributed sensing technology for monitoring greenhouse gas migration through vadose zone to the atmosphere," in *AGU Fall Meeting Abstracts*, vol. 2022, 2022, H45J–1511.

Books and Chapters

1. K. Araica, G. Cicala, M. P. McLeod, K. Soga, M. Jasiak, P. Hubbard, C.-C. Wang, **Y. Yang**, L. Luo, A. Yeskoo, et al., "Design and installation of distributed fiber optic sensing technology to monitor hdpe pipelines at an earthquake fault crossing," in *Pipelines 2024*, 2024, pp. 561–571.

Patents

1. *Flying driller: Gravity- and inertia-driven penetration with seed-inspired flight behavior for enhanced aerial seeding*, U.S. Provisional Patent Application No. 63/877,853, Sep. 2025.

Under Review

1. J. Gao, Y. Niu, **Y. Yang**, L. Luo, K. Soga, and C. Wang*, "Embedded fiber-optic sensing in printed concrete for structural failure and fracture monitoring," (Under Review at *Additive Manufacturing*), 2026.
2. M. Jasiak, **Y. Yang***, C.-C. Wang, S.-H. Chiu, L. Luo, P. Hubbard, Y. Xu, Y. Zhang, R. Burgmann, and K. Soga, "Transforming critical water pipelines into seismic fault movement sensors," (Under Review at *Geophysical Research Letters*), 2026.
3. L. Luo*, **Y. Yang**, K. Soga, J. Ryan, and Y. Wu, "Flow-induced vibration suppression for distributed fiber optic sensing cables in turbulent marine environments," (Under Review at *Journal of Ocean Engineering and Science*), 2026.
4. G.-B. Shin, **Y. Yang***, L. Pelecanos, F. A. Mevoli, E. Ouyang, and K. Soga, "A semi-analytical framework for thermal integrity profiling of cast-in-situ piles using temperature response functions," (Under Review at *Journal of Geotechnical and Geoenvironmental Engineering*), 2026.
5. Y. Wang, M. L. Delle Monache, **Y. Yang**, and K. Soga, "Optimal control modeling for switched systems in traffic evacuation," (Under Review at *Transportation Research Part C: Emerging Technologies*), Available at SSRN: https://ssrn.com/abstract=6727554, doi:10.2139/ssrn.6727554, 2026.
6. **Y. Yang***, T. Zhang, K. Kumar, K. Soga, C. M. Perez, L. Luo, and G.-B. Shin, "A matrix-free discontinuous galerkin method for dynamic cosserat rod simulation," (Under Review at *Computer Methods in Applied Mechanics and Engineering*; SSRN: https://ssrn.com/abstract=7417101), 2026.

7 S. Yi, Y. Hong, F. C. Jaeger, M. Freschi, **Y. Yang**, Z. Zhang, H. Lee, K. Soga, E. Moler, T. Zhang, T. L. Bauerle*, S. Yang*, and L. Yao*, "Hygrochastic seedpods: Engineering microclimates for ecological restoration," (Under Review at *Nature Ecology & Evolution*), 2026.

Professional Service

Peer Reviewer

- Journal of Geotechnical and Geoenvironmental Engineering (JGGE), ASCE
- Canadian Geotechnical Journal (CGJ), Canadian Science Publishing
- Applied Mathematical Modelling, Elsevier
- Computer Methods in Applied Mechanics and Engineering, Elsevier

Professional Memberships

- American Society of Civil Engineers (ASCE), Member
- American Geophysical Union (AGU), Member

Graduate Student Mentoring

- Maksymilian Jasiak, Shih-Hung Chiu, Yan Wu, Jiahui Yang, Fanbao Cheng

Research Projects

- **NSF GCR: Rooted in Nature** **2024 – Present**
  Bioinspired Design of Sustainable Seeding Methods to Improve Forest Regeneration
  National Science Foundation, Growing Convergence Research (GCR) Award No. 2428641
  University of Pennsylvania, University of California Berkeley, Syracuse University
  Role: Numerical simulation of soil mechanics and multiphase poromechanics; sensing design and numerical simulation of bioinspired soft actuators; developed DFOS FiberNet, a woven-structure and soft-robotics-inspired single-fiber tactile skin prototype.

- **NSF–UKRI SitS: Dynamic Coupling of Soil Structure and Gas Fluxes** **2019 – 2024**
  Measured with Distributed Sensor Systems: Implications for Carbon Modeling
  National Science Foundation / UK Research and Innovation, Signals in the Soil (SitS) Award No. 1935551
  University of California Berkeley, Colorado School of Mines, Rothamsted Research
  Role: Development of distributed sensor systems; sensing system design and signal processing for soil structure and gas flux monitoring.

- **CEC: Integrated Monitoring of Cetacean and Ocean Environmental Impacts** **2023 – Present**
  from Floating Offshore Wind Development on the Pacific Coast
  California Energy Commission (CEC) Award No. CEC-EPC-23-002
  Lawrence Berkeley National Laboratory, University of California Berkeley, Monterey Bay Aquarium Research Institute, Optasense
  Role: High-fidelity numerical simulation of mooring lines and underwater rope structures; development of efficient solvers for acoustic wave propagation under realistic oceanographic conditions.

- **DOE GTO: GeoSecure-RTS — Geothermal Security with Real-Time Simulation** **2026 – Present**
  U.S. Department of Energy (DOE), Office of Geothermal Technologies (GTO)
  Role: High-fidelity geothermal model development, calibration, and parametric studies; model-order reduction, adaptive meshing, and parallel efficiency improvements; development and application of novel computational schemes.

- **USACE / FEMA: Embankment Dam Safety — Cracking and Seismic Limitations** **2025 – Present**
  Levee Safety Protocols, Seismic Risk, and Embankment Cracking Assessment
  U.S. Army Corps of Engineers (USACE) & Federal Emergency Management Agency (FEMA)
  In collaboration with the USACE Dam Safety Production Center (DSPC-T)
  Role: Phase-field modeling of crack propagation in embankment soils; technical review and coordination on seismic risk and levee safety assessment.

Talks and Workshops

- **UC Berkeley Geotechnical Engineering Research Symposium** **2024**
  Keynote Speaker: "Fiber-Optic Shape Sensing and Error Analysis with the Cosserat Theory of Rods"
  Berkeley, CA
- **Hybrid Signals in the Soil (SitS) Workshop** **2024**
  Poster Speaker: "Locating Cracks Using Distributed Fiber Optics Sensing in Mesocosms and Field by Utilizing a Spectrum Based BOTDA"
  Alexandria, VA
- **NSF GCR PROJECT EVENT PROGRAM - Rooted in Nature: Bioinspired Design of Sustainable Seeding Methods** **2026**
  Keynote Speaker:"Mechanisms for Falling Seeds and Numerical Simulations"
  Berkeley, CA
- **International Joint Research Center for Resilient Infrastructure (ICRI) - Youth Lecturer Series** **2026**
  Keynote Speaker:"Advancing Distributed Fiber-optics Sensing through Physically Grounded Mathematical Frameworks"
  Shanghai, China

Teaching Experience

Fall 2019 — **Applied Geophysics**, Graduate Student Instructor (GSI/TA).
Received excellent teaching evaluations: **4.64/5.00** (Department Average: 4.23).

Fall 2021, 2024 and 2025 — **Infrastructure Sensing and Modeling**, Course Project Leader.
Led semester-long course projects integrating sensor data and modeling techniques.

2025 – Present — **Research Mentorship**, Postdoctoral Researcher.
Co-advising PhD students and junior researchers in Soga Research Group.

Skills

Languages	English (Fluent), Mandarin Chinese (Native), Russian (Basic).
Coding	C++, Python, Matlab, LaTeX.
Software	ABAQUS, COMSOL, AutoCAD, DealII.
Design	CAD, Tunnel Design.

Honors and Awards

- **National College Mathematics Competition** **2016**
  National Award: "First Prize (CMC - non Math major)"
  China
- **Tongji University Academic Excellence** **2016 – 2017**
  Scholarships: "First-Class Scholarship (2 Years)"
  Shanghai, China
- **Mathematical Contest in Modeling (MCM/ICM)** **2018**
  International Award: "Meritorious Winner (M Prize)"
  USA / International

Referees

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